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Understanding the Language of ADHD and Autism Communities on Social Media

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***Abstract*—Health communities online are popular for individuals to discuss health challenges and exchange social support. With social media, online communities also benefit neurodivergent individuals, by creating inclusive spaces where sharing of experience and knowledge is encouraged. The discussion in online communities covers a wide range of topics. As a result, the discussions differ in terms of topics, tone, and approach. This paper presents an analysis of social media posts shared on Reddit communities on Attention Deficit Hyperactivity Disorder (ADHD) and Autism Spectrum Disorders (ASD) between 2018 and 2020. In the study, we use a computer-aided model to identify prevalent topics in each subreddit and common themes. We conduct a comparative analysis of the communities and assess theme frequency and sentiment. The study highlights common topics found in r/adhd and r/autism subreddits, including diagnosis, treatment (medication dose and side effects), and social aspects (school, work, and peer interactions).**

I. Introduction

Social media provides a platform for users to share and seek medical information. Hence, by analyzing users' posts, researchers can characterize online communities, gaining insights about community responses to any medical concerns and diseases' prevalence and diagnosis [1]–[5]. Among various social media channels, Reddit's discussion forums[1] stand out as a popular social news website, covering topics ranging from news and politics to healthcare and jobs. The discussion on Reddit often involves sensitive topics that individuals feel uncomfortable to discuss in person or on websites where they can be personally identified [6]. Social media analysis is beneficial since it reaches a large number of users quickly, without requiring in-person interventions [7]. Consequently, analyzing social media, particularly Reddit, is a valuable tool that complements traditional user studies in the fields of Human-Computer Interaction and Social Computing [8].

There is an increasing amount of research dedicated to explore how social media activities can be used to analyze and improve mental health [8]. Reddit, which allows users to post anonymously, actively engages users in sensitive discussions on topics, including mental health, treatment, and diagnoses. Reddit also provides researchers with large datasets that can be used for social media analysis through natural language processing (NLP), qualitative analysis, and machine learning models [9], [10].

[1]www.reddit.com

Neurodivergent conditions impact the mental health, well-being and quality of life of individuals affected [11]. By engaging in social media channels, users are able to disclose mental health issues and seek support networks where they can share personal experiences. This engagement helps to reduce the stigma related to mental health. Reddit offers an open and anonymous platform for users to share concerns without fear of judgment or social stigma. As a result, the content shared on Reddit is of a high quality being more legitimate and less biased when compared to content obtained through surveys and interviews in which individuals are reluctant to share personal experiences and hesitate to disclose sensitive information [12].

Although mental health and neurodiversity have recently gained more attention, there is still a significant gap in understanding the specific language and communication patterns within online neurodivergent communities [13], [14]. To bridge this gap, our study aims to understand the nuances of how individuals with neurodivergent conditions, specifically ADHD and Autism, express themselves and seek support on Reddit. We focus on analyzing the content, tone, and themes prevalent in these forums to uncover insights into the lived experiences and challenges faced by these community members. This approach helps in understanding the real-world impact of neurodivergent conditions on individuals and offers a more empathetic perspective towards their daily struggles and coping mechanisms.

This paper reports the results of an analysis of Reddit posts published in ADHD and Autism forums. Our contributions include: (1) a computer-aided model to identify the most popular topics for each subreddit; (2) a list of common topics discussed in the communities, providing a comparative analysis of the emergent themes and contents shared in each community; (3) a study of the frequency and sentiment associated to each theme. The results show that common topics involve the diagnostics, the treatment (medication dose and side effects), social and community aspects (including school, work and interaction with peers). Our main contributions are driven by two research questions:

RQ1: To what extent social media analytics facilitate the understanding of online communities around ADHD and Autism?

RQ2: How similar are the themes discussed between these two communities?

II. Related Work

Researchers have investigated the potential of social media for vulnerable communities. Reddit, due to its anonymity feature, stands out as an information source for psychology research since it promotes discussion on sensitive topics [6], [8], [15], [16]. In the context of online communities, prior work has analyzed posts related to depression, anxiety and bipolar disorder [17]–[19]. Also, works on social media analytics focused on the language used by community members with ADHD [20], [21]. Park et al. (2018) examined similarities and differences in three communities including depression, anxiety, and post-traumatic stress disorder [22]. Using topic modeling, Gauthier et al. (2022) analyzed online communities such as r/stopdrinking and r/OpiatesRecovery, dedicated to substance use disorder recovery. Recovery communities allow individuals to share their experiences, seek support, and receive advice from peers [6].

Previous studies examined the relationship between ADHD and ASD [23]. ASD and ADHD are neurodevelopmental disorders that share similar traits, such as difficulties paying attention, impulsivity [24]. As a result, it is important to compare, contrast, and characterize the overlapping of discussion topics on social media in these communities. Insights gained from identifying similarities and differences in topics discussed within these communities can assist in identifying relevant areas for further intervention and support for individuals with ADHD and Autism. While Latent Dirichlet Allocation (LDA) and similar methods have advanced our understanding of online mental health discussions [25], [26], there is still room to explore the unique challenges and experiences faced by individuals living with conditions that share similar symptoms. In context of neurodiversity and social media, there has not been enough research on the overlapping traits between ADHD and Autism. In our study, the main goal is to identify the main themes and topics discussed within r/ADHD and r/Autism subreddit, and to investigate the nature of differences and similarities between ADHD and Autism subreddits.

III. Methods

A. Data Description

We analyze an existing dataset (see [27]) that includes posts from 15 of the world's largest mental health support groups, specifically focusing on the "r/autism" and "r/adhd" subreddits on Reddit. These posts were created between 2018 and 2020, with the r/ADHD subreddit contributing 45,631 posts and the r/Autism subreddit contributing 8,869 posts. This dataset offers a detailed view of the discussions and topics prevalent in these online communities during this period.

B. Model Description

Topic modeling provides structured insights by uncovering the thematic framework within large text datasets. In our study, we employ the LDA [25], a widely-adopted probabilistic topic modeling method. LDA posits that documents are compositions of multiple latent topics, with each word in a document attributed to one of these topics. This attribution facilitates the extraction of dominant themes pervasive across the document corpus. We utilized the Gensim toolkit [28] for the implementation of topic modeling, designed expressly for text processing tasks. Additionally, the Mallet toolkit [29], which employs Gibbs sampling, was adopted to enhance the efficiency of managing large-scale text corpora.

The methodological objective is to calculate a topic probability distribution for every word within a document. This calculation is informed by inputs such as the number of documents M, the number of topics t, and the vocabulary matrix β. Initially, a distribution over topics, parameterized by Alpha (α), is chosen for each document. Subsequently, every word W in a document d is probabilistically assigned to one of the t topics, based on the product of two probabilities: the probability of topic t given document d $P(Topic\, t \mid Document\, d)$, and the probability of word W given topic t $P(Word\, W \mid Topic\, t)$. The choice of word W for a topic t is influenced by the word's distribution in the vocabulary matrix β.

This procedure begins with a document-term matrix input, where each cell denotes the frequency of a term in a document. Through Gibbs sampling, the word-topic assignments are iteratively updated, thereby incrementally enhancing the models accuracy. This iterative refinement is continued until convergence is achieved, solidifying the reliability of the topic distribution estimations within the document set. The outcome is a set of topic probabilities for each document and a list of words most representative of each topic, exemplified in Table II and Table I. In this study, we maintained the default setting of the Alpha (α) hyperparameter in the Gensim implementation.

C. Analysis

First, we applied the LDA model to each subreddit using content analysis. Then, we examined each topic qualitatively and quantitatively, categorizing them based on their semantic content. This allowed us to understand the meaning and context of the conversations within each subreddit. Lastly, we estimated the sentiment of each topic. Also, we determined how similar the topics of two subreddits were based on their word lists and coherence.

1) Topics and sentiments:

Data Preprocessing: The data was preprocessed, including noise removal, error correction, and normalization, culminating in 49,677 words extracted from the ADHD set and 24,053 from the autism set. Our approach involved seven key preprocessing steps applied to our corpora: removing user tags and stopwords, replacing abbreviations and slangs, lemmatization, cleaning (correcting spelling and removing junk symbols), and removing hyperlinks. Each post in these datasets was then considered an individual document for topic analysis.

Determining Number of Topics: We conducted topic modeling using the LDA model with the Mallet software [29]. The optimal number of topics for LDA lacks a universal standard. However, a meta-analysis suggests that most studies select between 10 and 50 topics, with an average of 35 [30].

In our study, we evaluated two metrics: coherence (semantic relation within a topic) and exclusivity (uniqueness of words to that topic). Based on these metrics, we conducted an iterative analysis to determine the most suitable number of topics for our dataset. LDA was run iteratively, modeling every 5 topics from 10 to 35. The Autism model had the best coherence with 20 topics, while the ADHD model peaked at 35 topics. However, after qualitative assessment, we chose the ADHD model with 20 topics.

Qualitative Assessment: The intelligibility of models was examined based on topic word lists and coherence scores. Some issues were observed, like duplication of topics or topics that lacked clear thematic meaning. For instance, in an Autism model with 35 topics, there were two topics with overly similar words. Models with fewer than 20 topics seemed to merge distinct topics. Some topics were also excluded if they were not consistent with the post content. The result of this analysis was a list of words representing each identified topic.

ADHD Medication Extraction: Table I indicates the discussions surrounding medication, dosage, and potential side effects as prominent topics in ADHD-related posts. This suggests a keen interest from individuals with ADHD and their close ones in ADHD medications, dosages, and possible side effects. Hence, we conducted a separate analysis on this specific topic in the ADHD subreddit. Using regex, we identified mentions of FDA-approved ADHD medications. The top four, based on post mentions, were highlighted, shedding light on the most discussed ADHD medications.

Sentiment Analysis: We analyzed the characteristics of posts regarding their sentiments using the NLTK sentiment analysis toolkit [31]. Sentiment analysis helps determine the emotional tone of a text, such as an online post or a customer review. The NLTK analyzer assigns a compound score on a scale ranging from -1 to 1, which we used to categorize the posts as negative or positive. Then, we calculated the percentage of positive and negative posts in each cluster. These findings provide insights into the emotional experiences of those with ADHD and Autism. However, sentiment scores' interpretation vary with context; we focused on general emotional tones in posts.

2) Thematic overlap and similarity between subreddits:

Qualitative Method and Visualization: To determine similar contents shared in both subreddits, we used a qualitative method and evaluated our analysis with coherence scores. A qualitative analysis was conducted based on the most used terms for each topic and the content of several randomly selected posts. Additionally, to assess the overlap among topics, we applied word vector similarity across all pairs of topics. A word vector is a mathematical representation of a word, capturing its context and meaning numerically. This allows for the quantification of similarity between words and documents, as well as between topics, within a multidimensional space. In our study, we used a document vector similarity analysis to each pair of topics, one from each subreddit. This revealed how similar or dissimilar the topics are, and whether there are overlaps between them. We used 'en_core_web_md', a set of pre-trained word vectors from spaCy [32]. Using LDA, we identified the topic with the highest percentage of contribution in each post and assigned it as the main topic for the post.

The similarity between two sentences is determined using the "dot product" and normalization of vectors. We selected the top 50 posts based on percentage contribution per topic and measured the similarity between pairs of posts from different topics. Each post was assigned an average similarity score, and an overall average was calculated.

IV. Results

A. RQ1: Primary Themes in Reddit Communities

ADHD Themes:

1. **Medication (n=4,781):** Reddit users shared their experiences with ADHD medications, including Adderall (1,692 mentions), Ritalin (570 mentions), Vyvanse (1,092 mentions), and Concerta (613 mentions). Feedback covered both positive and negative effects, highlighting therapeutic benefits and potential side effects such as anxiety, appetite loss, weight reduction, and insomnia. Se also identified posts that discuss specific side effects, including sleep and insomnia, anxiety, nervousness, restlessness, irritability, crash, fatigue, tiredness, appetite, cravings, and weight loss, using keywords related to side effects. We considered only posts that mentioned a single medication and not those that mentioned two or three brands at the same time. Moreover, we analyzed the frequency of side effect mentions in relation to each brand. As shown in Fig. 1, the highest number of mentions of Vyvanse was for sleeping problems, Ritalin for fatigue, and Concerta for anxiety and nervousness. Overall, our analysis suggests concerns regarding the most commonly prescribed brands for ADHD symptoms and specific side effects frequently associated with these brands.

2. **Memory Problem (n=3,584):** Posts discussed memory challenges, especially short-term memory lapses. Individuals mentioned misplacing items, forgetting tasks, and missing important dates. Users proposed strategies like designated item locations, calendars, and breaking tasks into smaller steps to cope.

3. **Sleep and Eating Habits (n=3,017):** Topics included difficulty remembering to eat, sleep problems, and loss of appetite due to stimulant medication. Suggestions included creating schedules, diet plans, exercise, therapy, weighted blankets, and relaxation exercises.

AUTISM Themes:

1. **Sensory and Noise Problem (n=509):** Individuals with autism spectrum disorders discussed sensory sensitivities and noise-related challenges. Recommendations for noise-canceling headphones and strategies for coping with sensory overload and clothing sensitivities were shared. Accommodations like noise-canceling headphones and alternative clothing options were suggested.

2. **Therapy (n=447):** Topics covered speech therapy, ABA therapy, challenges faced by nonverbal children with ASD, and managing behavioral issues. Parents shared experiences and sought advice on communication and relationships.

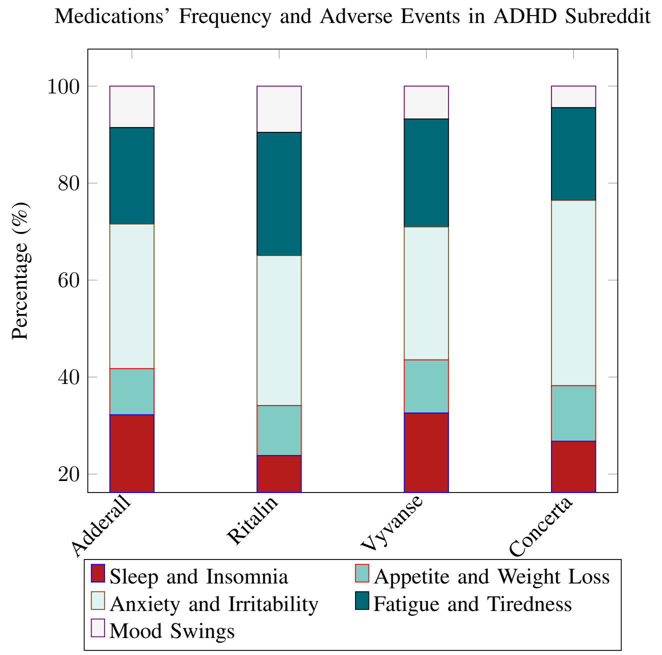

Figure 1: Medications' Frequency and Adverse Events in ADHD Subreddit (Normalized)

3. **Friendship (n=465):** Posts addressed social interactions, behavior's impact on relationships, and seeking advice to improve friendships. Challenges included initiating conversations, interpreting communication patterns, self-worth, and meltdowns. Effective communication, patience, and support systems were highlighted as important for individuals with ASD in forming and maintaining friendships.

B. RQ2: How similar are the themes discussed between the communities?

Semantic similarity analysis reveals a significant overlap in topics discussed in the Autism and ADHD communities. Five topics with the same labels show high similarity on the heatmap (Fig. 3, confirming that our manual labeling accurately captures the themes in these topics. We used a similarity metric to assess the overlap in topic content, with average scores visualized in a heatmap (Fig. 3 in appendix). Analysis of posts identified four common themes across both communities: work-related concerns, diagnoses, social interactions, and family/school issues, depicted in a Venn diagram Fig.2

C. Themes in Autism and ADHD Communities

Work: Both the Autism and ADHD communities express concerns about employment. Individuals with Autism face difficulties in finding and keeping jobs, communicating with coworkers, and discrimination post-disclosure. They also mention challenges with inconsistent schedules and transitions. The ADHD community shares similar concerns about employment but also emphasizes productivity, focus, and time management.

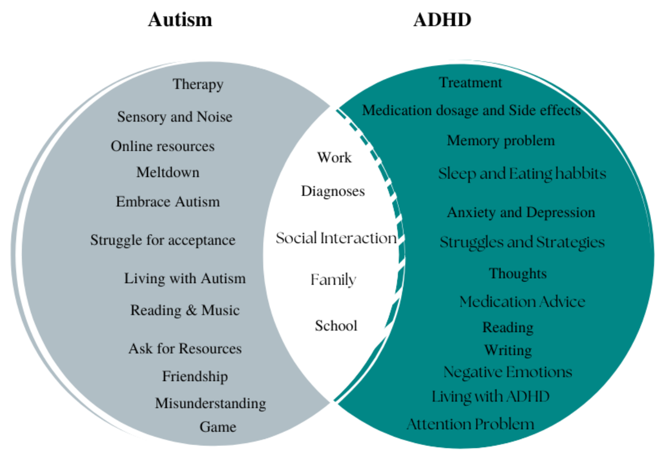

Figure 2: Topics with highest thematic overlap.

"I would like to know if there are any special rules or laws in regards to being fired after disclosing my autism to my supervisor." - Reddit User

Diagnoses: Posts in this topic discuss issues related to informal or self-diagnosis. The importance of accurate diagnosis is emphasized, and challenges arise when individuals disagree with their diagnosis. The diagnostic process relying on self-reported symptoms is a common concern.

Family: In the Autism community, posts revolve around experiences with Autism within family dynamics. Users seek advice on supporting autistic family members under societal pressures and worry about family perceptions post-diagnosis. The ADHD community discusses the experiences of individuals with ADHD and their relationships with their family members, highlighting misunderstandings and the need for education.

"I still struggle with talking with other family members other than the ones that live with me." - Reddit User

School: Posts on r/ADHD highlight educational challenges faced by those with ADHD, including difficulties in focusing, procrastination, impulsiveness, and organizational issues. In r/Autism, the importance of individualized education plans, appropriate classroom placement, and accommodations for children with autism is emphasized.

"I hate how people perceive ADHD. When people think of ADHD, they think: 'oh that kid can't do their homework'." - Reddit User

Social Interaction: In r/ADHD, individuals describe struggles in social interactions, including maintaining friendships, feeling out of place in group conversations, understanding and managing empathy, and forming romantic relationships. Users also report trouble paying attention in conversations and self-consciousness. *"I feel like every time I talk to anyone, I'm in my head and self-conscious of everything I had just said." - Reddit User*

In r/Autism, users share experiences with difficulties recognizing sarcasm, language and communication challenges, and discomfort with eye contact.

"I find myself struggling with eye contact. Looking people in the eyes for more than a moment or two makes me quite uncomfortable." - Reddit User

D. Sentiment Analysis

The sentiment analysis results provide an overview of the sentiment expressed in the dataset related to ADHD and Autism. From the results, we can conclude that: most topics received a relative balanced number of positive and negative sentiments, with a higher percentage of positive sentiments than negative ones. The topics of Entertainment, Reading, and Living with ADHD, have the highest percentage of positive sentiment, indicating that these topics are generally discussed in a positive way.

The topics like Anxiety, Depression, Feelings, Family, Thoughts, School, Treatment, Memory Issues, Sleep/Eating Habits, Diagnoses, Symptoms, Work, Attention Problems, Medication (Advice, Dosage, Effects), and Social Interaction in the study showed a lower positive sentiment percentage. This suggests these topics often contain posts with negative sentiments (refer to Fig.4a in appendix for details).

The results of sentiment analysis revealed a diverse range of sentiments across various topics related to autism. The topic of Online resources had the highest percentage of positive sentiment at 84.98%, while meltdown had the highest percentage of negative sentiment at 37.17%. The topics of Therapy, Reading & Music, Asking for Resources, Games, Friendship, and School had a high percentage of positive sentiment ranging from 81.12% to 69.08%. The topics of Getting Advice, Family, Social Interaction, Work, Embrace Autism, Diagnoses, and Symptoms, Living with Autism, Misunderstanding, Sensory and Noise problem, Struggle for Acceptance, and Meltdown had a relatively lower percentage of positive sentiment ranging from 64.18% to 55% (see Fig. 4b in appendix).

V. Discussion

This study conducted an analysis of Reddit posts to delve into the discourse of neurodiversity on social media. While these findings may have implications across various social media platforms, it is essential to recognize the need for further research to validate their applicability. Also, limitations in the study, such as using a single post per user and discrepancies in dataset sizes between r/ADHD and r/Autism subreddits, should be acknowledged, prompting the necessity for more extensive investigations and potential impact on result representation. Nevertheless, the study's methodology offers a promising approach for remote ethnographic studies and broader participant inclusion in neurodiversity-related research.

A notable limitation of this study is the inherent challenge in verifying the authenticity of each user's diagnosis. Given the anonymous nature of Reddit, there is an underlying assumption regarding the conditions of users, which may not always be accurate. Additionally, the presence of allies, such as parents and caregivers, who may not personally have ADHD or Autism but participate in discussions, further complicates the interpretation of data. The blend of direct and indirect experiences enriches the discourse but also adds a layer of complexity to the analysis. The study's methodology offers a promising approach for remote ethnographic studies and broader participant inclusion in neurodiversity-related research.

Additionally, the study highlights the active discussions surrounding ADHD medication, and its impact on various aspects of life such as social interactions, family dynamics, educational and occupational challenges, and emotional responses. The research underscores the potential for future inquiries into adverse events associated with ADHD medication and emphasizes the importance of considering individuals with ADHD's perspectives when examining medication effects. As individuals with ADHD share their experiences and concerns regarding medication on social media, researchers have a valuable resource for studying this topic comprehensively. Still, the study findings must be interpreted within the context of a social forum open to the public neither including nor serving as medical advice. Study findings provide valuable insight into the lived experiences of individuals with ADHD and ASD from a first-person or ally perspective.

VI. Conclusion & Future Work

This study explored the discussion themes on two popular subreddits for individuals with ADHD and Autism, using a combination of quantitative and qualitative analysis methods with different visualizations. The analysis revealed common concerns spanning diagnosis, treatment (including medication dose and side effects), and social aspects (such as school, work, and peer interactions) shared by both communities. Notably, sentiment analysis indicated a positive tone in posts by neurodivergent adults, although individuals with ADHD expressed more negativity on certain topics. These findings emphasize the significance of considering the viewpoints of individuals with ADHD and Autism when researching related subjects. Moreover, the use of social media platforms as data sources holds promise for future research on neurodivergent communities, offering valuable insights that can inform more comprehensive studies of their experiences and concerns.

In our study, we employed a LDA, traditional topic modeling method, which views documents as mixtures of multiple topics, each defined by a distinct word distribution. Although LDA has proven effective, it can sometimes lead to topic representations that include irrelevant or incoherent words. To address this limitation, we are exploring the use of Large Language Models (LLMs) for their advanced analytical capabilities [33]. Initial results, aligning with previous work, reveal that while LDA offers simplicity and interpretability, LLMs provide more detailed and contextually rich insights into topics, indicating their potential for deeper analysis in topic modeling [34]. Mindful of the potential biases associated with LLMs [35], we recognize the importance of ethical considerations in their use. Moreover, digital ethnography focused on underrepresented groups, such as neurodivergent communities, can greatly benefit from LLMs as these models are capable of revealing deeper insights and patterns in social media discussions that traditional methods may overlook.

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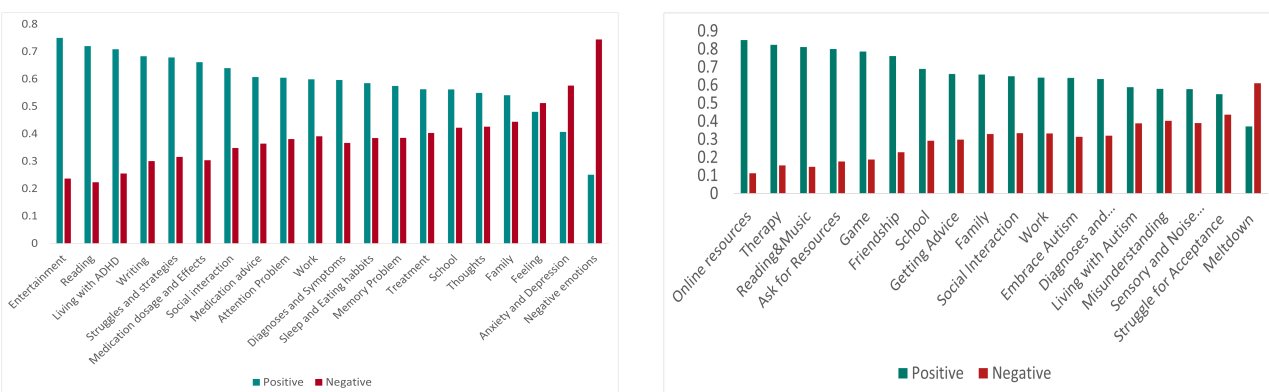
Appendix

Figure 3: **Word-Vector-based document semantic similarity** between pairs of topics from each subreddit.



(a) Topic-based sentiment analysis for ADHD

(b) Topic-based sentiment analysis for Autism

Figure 4: Sentiment Analysis for ADHD and Autism

Table I: ADHD Topics

Topic	Keywords	Frequency
Medication dosage and Effects	Adderall, mg, Vyvanse, taking, day, dose, Ritalin, effects, Concerta, prescribed	4781
Memory Problem	phone, put, room, home, forgot, house, forget, remember, memory, brain	3584
School	school, college, class, year, high, study, classes, studying, grades, semester	3386
Sleep and Eating habbits	day, sleep, hours, morning, night, days, eat, bed, hour, tired	3017
Diagnoses and Symptoms	adhd, diagnosed, diagnosis, symptoms, adult, therapist, disorder, recently, psychiatrist, treatment	2883
Therapy and Treatment	doctor, appointment, psychiatrist, told, insurance, month, prescription, months, medication, finally	2759
Entertainment	things, thing, find, love, time, music, video, watch, play, games	2424
Social interaction	people, adhd, friends, talk, person, talking, lot, social, understand, friend	2335
Anxiety and Depression	anxiety, depression, experience, issues, anxious, heart, strattera, mood, similar, worse	2191
Thoughts	brain, mind, head, thoughts, thinking, stop, thing, thought, focus, times	2103
Reading	amp, ADHD, read, reading, post, HTTPS, found, Reddit, book, find	2082
Work	work, job, time, working, home, money, good, jobs, make, career	2035
Family	adhd, years, life, parents, told, family, mom, diagnosed, year, kid	1869
Attention Problem	things, time, work, focus, hard, start, tasks, task, distracted, lot	1763
Negative emotions	shit, hate, make, thing, fucking, bad, dont, time, stupid, fuck	1601

Table II: Autism Topics

Topic	Keywords	Frequency
Diagnoses	diagnosed, autism, diagnosis, spectrum, years, symptoms, doctor, ADHD, anxiety, adult	784
Sensory and Noise problem	sensory, issues, loud, sound, noise, stimming, hands, sounds, stim, lot	680
Therapy	son, child, kids, year, therapy, children, aba, daughter, speech, behavior	610
Friendship	people, social, friends, friend, make, talk, person, lot, close, anxiety	609
Embrace Autism	autism, autistic, people, person, neurotypical, show, understand, subreddit, spectrum, character	509
Online resources	autism, HTTPS, research, www, interested, study, experiences, share, form, online	493
Struggle for Acceptance	feel, people, things, normal, bad, feeling, hate, don't, makes, thing	460
Meltdown	back, stop, meltdown, meltdowns, start, started, point, room, head, crying	452
Getting advice	functioning, advice, high, wondering, things, tips, guys, thing, good, food	427
School	school, high, year, class, college, started, special, teacher, teachers, learning	427
Social Interaction	time, understand, hard, contact, eye, words, speak, lot, thing, talk	406
Work	work, job, working, time, find, money, jobs, worked, due, phone	402
Living with Autisms	day, time, days, today, sleep, night, hours, week, months, home	395
Family	family, parents, brother, mom, years, live, mother, life, advice, dad	379
Misunderstanding	things, told, talking, wrong, talk, asked, thought, explain, stuff, answer	360